Uncertainty-Aware Noisy Label Learning for Remote Sensing Change Detection

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***Abstract*—Deep learning (DL)-based methods have achieved tremendous success in remote sensing (RS) change detection (CD). However, most DL-based methods heavily rely on high-quality labeled samples, where noisy labels are inevitably introduced during the annotation process, especially at edge regions. Under the supervision of such noisy labels, the performance of RS CD models will deteriorate significantly. To break the limitation, a novel uncertainty-aware noisy label learning network (UNLLNet) is proposed for RS CD, termed UNLLNet. Specifically, a joint detection strategy based on uncertainty analysis is proposed, which leverages samples characterized by low uncertainty and high probability to detect and correct potential noisy labels, thereby mitigating the risk of erroneous correction. Given that noisy labels are more likely to be introduced at edge regions, an edge-guided interlevel difference refinement module (EIDRM) is designed to effectively calibrate the edge structure of change regions, thereby facilitating better identification of noisy labels at edge regions. Moreover, a noise-robust loss function with adaptive hard sample enhancement is introduced, assigning loss weights to each sample based on their uncertainty to further enhance the model's robustness over noisy labels. Experimental results on LEVIR-CD, CDD, and CLCD datasets validate the effectiveness and advantages of UNLLNet compared to other state-of-the-art RS CD methods.**

***Index Terms*—Edge-guided interlevel difference refinement, joint detection strategy, noise-robust loss function, noisy labels, remote sensing (RS) change detection (CD).**

NOMENCLATURE

DL	Deep learning.
RS	Remote sensing.
CD	Change detection.
HR	High-resolution.
MLP	Multilayer perception.
UNLLNet	Uncertainty-aware noisy label learning network.
EIDRM	Edge-guided interlevel difference refinement module.
CNN	Convolution neural network.
GT	Ground truth.
ILAtt	Interlevel attention.
KL	Kullback–Leibler.
CE	Cross-entropy.
F1	$F1$-score.
IoU	Intersection over union.
OA	Overall accuracy.
TP	True positive.
TN	True negative.
FN	False negative.
FP	False positive.
Params.	Parameters.
FLOPs	Floating point operations per second.
SOTA	State-of-the-art.

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I. INTRODUCTION

CD HAS emerged as one of the critical technologies in earth observation [1], [2]. It aims to identify changes within the same geographical area but in different periods. With the advancement of sensor devices, various types of RS images can be available for land resource management [3], urban planning [4], disaster assessment [5], [6], and environmental monitoring [7], etc. Among these images, HR RS image acquires tri-band electromagnetic waves radiated by ground objects through synchronous photographic or scanning methods [8]. It is capable of capturing fine-grained ground scattering information, thus offering a distinct advantage in CD tasks due to its enhanced spatial and spectral discriminability.

Over the past few decades, traditional CD methods have significantly advanced the field of RS applications [9], [10]. Despite their contributions, traditional methods generally rely on handcrafted features, resulting in limited capability in complex scenarios. Recently, the rapid development of DL technology has demonstrated its superiority across various fields [11], [12], [13]. RS CD methods based on a convolutional neural network (CNN) have achieved great success due to their powerful feature extraction and representation capabilities. For example, Daudt et al. [14] proposed three fully convolutional neural networks with different structures (e.g., FC-EF, FC-Siam-Diff, and FC-Siam-Conc). Zhou et al. [15] presented a joint frequency-spatial domain network, in which the frequency information was introduced into change features. However, the inherent constraints of a

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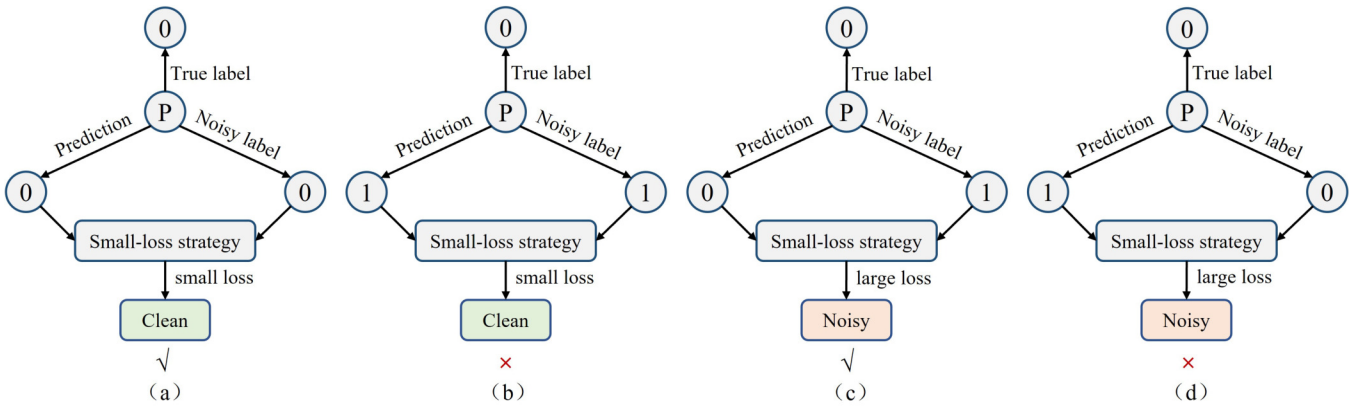


Fig. 1. Illustration of small-loss strategy. P is a pixel sample, with 0 and 1 indicating its class label. The alignment between prediction and true label indicates the correctness of the model's prediction, while the alignment between noisy label and true label indicates whether P is a noisy sample.

fixed receptive field in CNN models hinder their ability to model long-range dependencies [16]. To solve this problem, Chen et al. [17] introduced the Transformer into RS CD to better capture global features, which has gained popularity. Besides, a Transformer-based Siamese network architecture for RS CD was proposed in [18], which achieved accurate CD by rendering multiscale long-range details. Recent advancements have combined the strengths of CNN and Transformer [19], [20], [21], [22], or have turned toward diffusion models [23], [24], [25], [26] and Mamba [27], [28], [29], [30], achieving a significant progress in RS CD.

Despite the outstanding performance of these DL-based RS CD methods, they share a common limitation, i.e., a heavy reliance on high-quality annotations, which can be time-consuming and laborious to obtain [31]. During the process of data annotation, it is hard to precisely delineate the edge of change regions due to the lack of prior information or subjectivity in manual annotation [32], [33]. Noisy labels, which indicate the incorrect label information [34], are inevitably introduced and significantly impair the CD performance.

The noisy labels problem has attracted extensive attention from researchers; numerous methods have been proposed and shown promising results, which are summarized in Table I. These methods can be broadly classified into three types: transition matrix-based methods [35], [36], [37], [38], regularization-based methods [39], [40], [41], [42], [43], and sample selection/correction-based methods [44], [45], [46], [47], [48]. Transition matrix-based methods aim to estimate the transition probability between noisy labels and true labels (i.e., GT, which refers to the objective reference data obtained through precise measurements or expert annotations, serving as the benchmark for validating model outputs in machine learning). Jiang et al. [35] developed a random label propagation algorithm to estimate the transition matrix, finally reducing the level of noisy labels. However, it is difficult to accurately estimate the transition matrix in practical situations due to the failed prior assumption [45]. To be free of transition matrix estimation, regularization-based methods focus on designing robust loss functions or regularization techniques, exhibiting enhanced robustness to noisy labels. For instance, Liu et al. [39] designed an early learning regularization term to mitigate the interference of noisy labels on the optimization trajectory of the model. Zhou et al. [41] pioneered a novel loss function called asymmetric loss, which is robust to noisy labels due to its adherence to Bayesian optimization principles. Since all samples are utilized without correction during the training process, regularization-based methods may perform poorly under extremely noisy conditions. In contrast, sample selection/correction-based methods employ specific strategies to select possibly clean samples or correct noisy samples, among which the small-loss strategy is most widely adopted [49], [50]. This strategy classifies samples with small loss as clean and those with large loss as noisy, which subsequently will be discarded or corrected. For instance, Kim et al. [51] proposed a scheme for multilabel classification, which rejected or corrected the large loss samples during the training process, thereby preventing the model from overfitting to noisy samples. Han et al. [52] designed a DL paradigm called "Co-teaching," specifically, they trained two deep neural networks simultaneously, and the parameters of each network were updated under the guidance of small-loss instances from its peer network, which are considered more likely to be correctly labeled samples. However, these methods cannot distinguish between hard samples and noisy samples, as they all exhibit a large loss. As shown in Fig. 1(a) and (c), when the model prediction aligns with the true label, the clean sample exhibits small loss, while the noisy sample exhibits large loss. According to this, the small-loss strategy can effectively identify the noisy sample. However, once the model prediction is incorrect, the strategy may misclassify noisy sample as clean [illustrated as Fig. 1(b)], or conversely, misclassify clean sample (i.e., hard sample) as noisy [depicted as Fig. 1(d)]. Consequently, the small-loss strategy tends to erroneously categorize hard samples as noise, resulting in the erroneous correction.

Although significant progress has been achieved in handling noisy labels, these advancements are often unsatisfactory. In summary, the main challenges in the field of RS CD can be summarized as follows.

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TABLE I
GENERAL REVIEWS ABOUT RS CD AND NOISY LABELS

Reference	Year	Description
RS CD methods:		
Barber [9]	2015	Extending generalized likelihood ratio test for polarimetric sar change detection
Ciuonzo et al. [10]	2017	A unified testing framework for covariance matrix equality
Yang et al. [12]	2024	An untrained network for unsupervised hyperspectral image change detection
Daudt et al. [14]	2018	Fully convolutional Siamese networks for change detection
Zhou et al. [15]	2022	Integrating frequency information to enhance spatial feature extraction
Yu et al. [16]	2024	A mask classification-based change detection method for remote sensing images
Chen et al. [17]	2021	Remote sensing image change detection with transformers
Bandara et al. [18]	2022	A siamese network integrating hierarchical transformers and MLP decoder for accurate change detection
Liu et al. [19]	2022	Combining CNN-based feature extraction and transformer-based multiscale context aggregation for change detection
Liu et al. [20]	2023	An attention-based multiscale transformer network for remote sensing image change detection
Ji et al. [21]	2024	Domain adaptive and interactive differential attention network for remote sensing image change detection
Xing et al. [22]	2024	A hybrid cnn-transformer multitask network for joint open-pit mine extraction and change detection
Wen et al. [23]	2024	A generative change detection model based on difference-feature guided ddpm
Sun et al. [24]	2024	Merging feature extraction and distribution learning for remote sensing change captioning
Gedara et al. [25]	2022	Denoising diffusion probabilistic models as feature extractors for change detection
Jia et al. [26]	2024	A change detection model that combines discriminative and generative architecture
Zhang et al. [27]	2024	A novel network for enhanced remote sensing change detection in difficult cases
Liu et al. [28]	2024	Iterative mamba diffusion change-detection model for remote sensing
Chen et al. [29]	2024	Remote sensing change detection with spatio-temporal state space model
Wang et al. [30]	2024	Tree topology mamba-guided network collaborative hierarchical incremental aggregation for change detection
Yang et al. [31]	2024	Using difference augmentation and progressive reconstruction for hyperspectral change detection with limited labels
Transition matrix-based methods for handling noisy labels:		
Li et al. [32]	2020	Learning multiview CNNs and correcting error labels are alternatively conducted in an iterative manner
Wang et al. [33]	2023	An attentive-adaptive network that can be applied to both clean and noisy training sets
Kim et al. [34]	2024	Introducing instance-dependent multilabel noise using a vision-language model
Jiang et al. [35]	2018	A random label propagation algorithm is developed to cleanse the label noise.
Jiang et al. [36]	2020	A multiscale segmentation-based multilayer spectral–spatial graphs method for label noise cleansing
Luo et al. [37]	2024	Estimating per-class statistics for label noise learning
Liao et al. [38]	2025	Instance-dependent label distribution estimation for learning with label noise
Regularization-based methods for handling noisy labels:		
Liu et al. [39]	2020	A regularization-based framework that leverages early learning in deep neural networks
Zhou et al. [40]	2021	Learning with noisy labels via sparse regularization
Zhou et al. [41]	2023	Asymmetric loss functions for noise-tolerant learning
Zhou et al. [42]	2025	An in-depth analysis of the local effects of label noise on neural networks from a topological perspective.
Xu et al. [43]	2024	A class activation regularization network with class activation dropping and hard sample mining
Sample selection/correction-based methods for handling noisy labels:		
Xu et al. [44]	2020	A superpixel guided sample selection neural network for handling noisy labels in hyperspectral image classification
Han et al. [45]	2019	A robust deep self-learning framework for ConvNet training on real noisy datasets without extra supervision
Liu et al. [46]	2021	Noise-tolerant medical image classification via mutual label correction
Cordeiro et al. [47]	2023	Robust learning with high confidence samples in a noisy label environment
Zhu et al. [48]	2023	Robust co-teaching learning with consistency-based noisy label correction for medical image classification
Zhang et al. [49]	2024	A novel label noise type called BadLabel is introduced
Yue et al. [50]	2024	Clustering training losses for label error detection
Kim et al. [51]	2022	Novel methods which reject or correct the large loss samples to prevent model from memorizing the noisy label
Han et al. [52]	2018	A deep learning paradigm called “Co-teaching” for combating with noisy labels.

1) How to deal with the problem of noisy labels in RS CD? Existing methods inherently presume label perfection, neglecting the problem of noisy labels. However, the incorrect label information provided by noisy labels distorts the optimization trajectory of the model, significantly degrading the CD performance.
2) How to achieve more comprehensive exploitation of edge and change semantics to better identify the edge of change regions? This is of critical importance, particularly in noisy labels scenarios, as noisy labels are more likely to be introduced at edge regions. Most CD methods fail to extract sufficiently change-related edge information to handle label noise at edge regions.
3) How to mitigate the negative impact of undetected noisy labels? It is impractical to detect and correct all noisy labels for RS CD. However, undetected noisy labels may still impair the optimal training of the model.

To tackle the aforementioned challenges, a novel uncertainty-aware noisy label learning network is proposed for RS CD, termed UNLLNet. Specifically, an innovative joint detection strategy based on uncertainty analysis is proposed for noisy labels detection. By integrating the uncertainty and probability of each example, the joint detection strategy can select the most reliable label information to detect and correct potential noisy labels, and meanwhile mitigate the risk of erroneous detection that is common with the small-loss strategy. Aiming at fully exploiting edge and change semantics, an EIDRM is introduced, which facilitates comprehensive interaction and mutual reinforcement between

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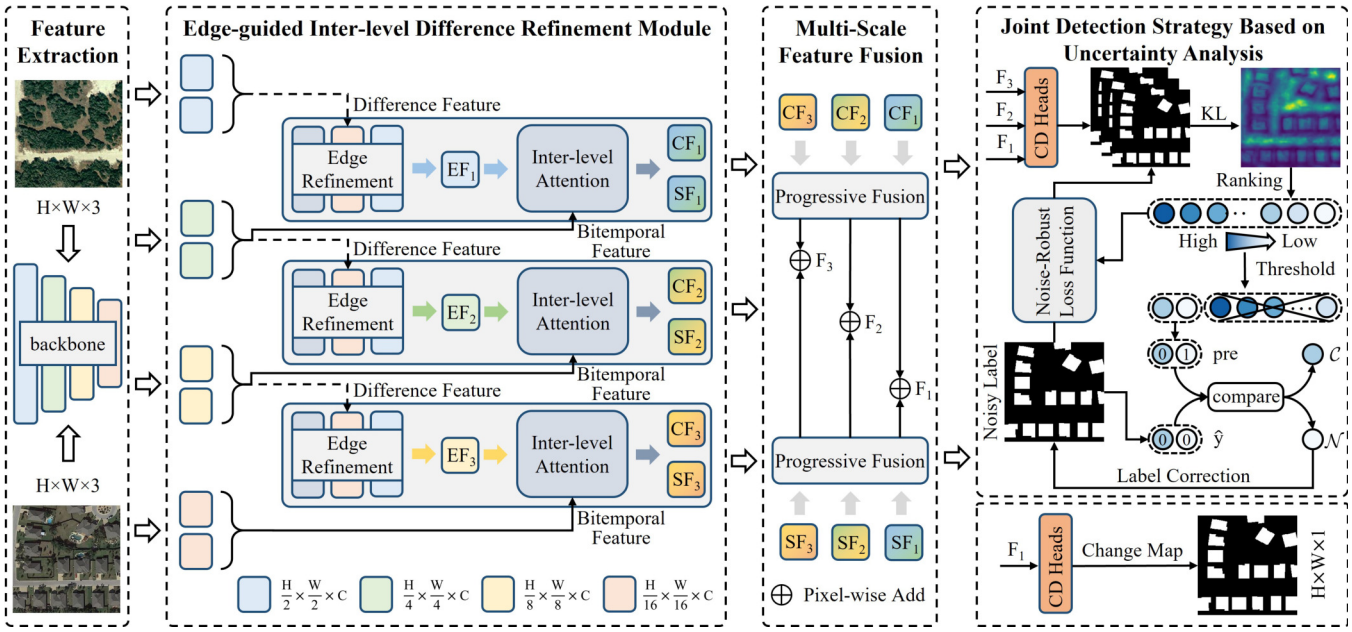


Fig. 2. Pipeline of the proposed UNLLNet. The overall network involves four parts: feature extraction, EIDRM, multiscale feature fusion, and a joint detection strategy based on uncertainty analysis. The final change map is obtained from F_1.

edge-enhanced low-level features and high-level bitemporal features, thereby calibrating the edge structure of change regions. Since EIDRM enables the model to generate better CD results, it implicitly improves the capability of UNLLNet to detect noisy labels. To further alleviate the negative impact of undetected noisy labels, a noise-robust loss function with adaptive hard sample enhancement is proposed, which adaptively assigns loss weights for each sample based on their uncertainty; that is, higher uncertainty indicates a harder sample and therefore results in larger loss weights, vice versa. In summary, the main contributions of this article are as follows.

1) An innovative method for the RS CD task with noisy labels is proposed, named UNLLNet, which pioneers a joint detection strategy based on uncertainty analysis to detect potential noisy labels by considering both uncertainty and probability. Especially, it can mitigate the risk of erroneous detection that is common with the most widely used small-loss strategy.
2) The EIDRM is introduced to fully exploit edge and change semantics through interaction between different levels, thereby calibrating the edge structure of change regions. In addition, the improved CD results facilitated by EIDRM implicitly enable the model to better detect noisy labels.
3) A noise-robust loss function with adaptive hard sample enhancement is proposed to further enhance the model robustness over noisy labels. By integrating the uncertainty of each sample into the loss function, the UNLLNet can achieve a superior fit for more complex and challenging feature representations.

The rest of this article is organized as follows. Section II describes the details of the proposed UNLLNet. In Section III, a comprehensive experimental evaluation is presented. Finally, the conclusion is given in Section IV.

II. Proposed Method

A. Architecture Overview

The overall architecture of the proposed UNLLNet is illustrated in Fig. 2, which comprises four steps: feature extraction, edge-guided interlevel difference refinement, multiscale feature fusion, and joint detection based on uncertainty analysis. The UNLLNet is described specifically below.

Given the bitemporal images T_1, $T_2 \in \mathbb{R}^{H\times W\times C}$, where H, W, and C represent height, width, and channel respectively, a Siamese weight-sharing backbone network (i.e., ResNet50 [53]) is first utilized to extract multiscale features, denoted as F_1^i and F_2^i, $i \in \{1, 2, 3, 4\}$.

The coarse-grained difference features, with spatial size of $(H/2^i) \times (W/2^i)$, are obtained from the first three feature blocks and fed into the designed EIDRM to fully exploit edge and change information. The key idea of EIDRM is to utilize the high-level features with rich change semantics to help locate the change-related edge in low-level features, achieving comprehensive interaction and mutual reinforcement between different levels. It implicitly bolsters the capability of UNLLNet to detect noisy labels. The refined difference-features output by EIDRM can be formulated as follows:

$$CF_i, SF_i = f_E\left(\left|F_1^i - F_2^i\right|, F_1^{i+1}, F_2^{i+1}\right) \tag{1}$$

where f_E represents the EIDRM. CF_{i} and SF_{i} are the difference features derived through concatenation and subtraction operations, respectively.

Following this, the refined difference features are aggregated to further enhance the changes by progressive upsampling. Specifically, CF_3 and SF_3 are first upsampled to align with

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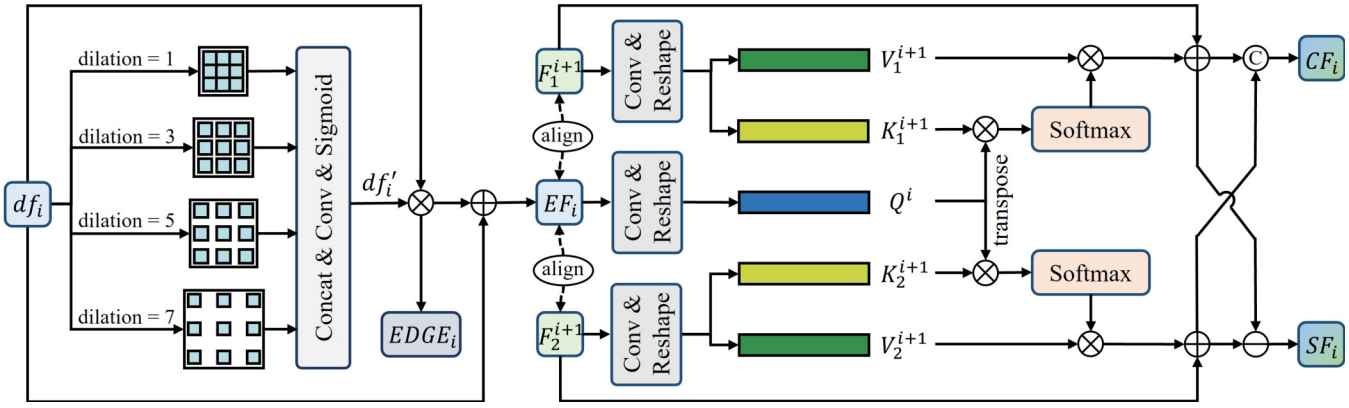


Fig. 3. Architecture of EIDRM. CF_i and SF_i are generated difference features derived through concatenation and subtraction operations, respectively.

the lower-level features and processed by a convolution block. The resulting features are subsequently concatenated with the lower-level features along the channel dimension. Ultimately, the multiscale features are progressively aggregated in the same manner. The entire process can be represented as follows:

$$F_{c1} = f_{3\times3}^1\left(\left(\left[f_{3\times3}^2\left(\left(\left[f_{3\times3}^3\left(CF_3\uparrow\right),CF_2\right]\right)\uparrow\right),CF_1\right]\right)\uparrow\right) \quad (2)$$

$$F_{s1} = f_{3\times3}^1\left(\left(\left[f_{3\times3}^2\left(\left(\left[f_{3\times3}^3\left(SF_3\uparrow\right),SF_2\right]\right)\uparrow\right),SF_1\right]\right)\uparrow\right) \quad (3)$$

where $\uparrow$ and $[\cdot]$ denote the bilinear upsampling and concatenation operation, respectively. $f_{3\times3}^i$ stands for a convolution layer with a filter size of 3×3, followed by batch normalization and nonlinear activation layers. The three intermediate features from the convolution blocks at corresponding levels are added together, and separately processed by three CD prediction heads f_{head}, where P_{cd1} is treated as the final CD results

$$\text{P}_{\text{cd}(i)} = f_{\text{head}}^i\left(F_{c(i)} \oplus F_{s(i)}\right). \quad (4)$$

A joint detection strategy based on uncertainty analysis is proposed to detect and correct potential noisy labels, and the corrected labels are then fed into the next iteration. With respect to the optimization of the network parameters, a noise-robust loss function with adaptive hard sample enhancement is introduced, which enables UNLLNet to generate high-quality CD results and, meanwhile, further enhances its robustness over noisy labels.

B. Edge-Guided Interlevel Difference Refinement Module

As acknowledged, the incorporation of edge structure information can effectively improve the blurred edge of CD results [54], [55]. Most existing RS CD methods cannot capture sufficient edge features, resulting the poor performance in complex change scenarios, particularly in the presence of noisy labels, as they are more likely to be introduced at edge regions. Therefore, as illustrated in Fig. 3, an EIDRM is proposed to fully exploit edge and change semantics. Specifically, the change semantics derived from the high-level features are utilized to locate change-related edges in low-level features. Through comprehensive interaction and mutual reinforcement between different levels, the designed EIDRM can effectively calibrate the edge structure of change regions and implicitly improve the capability of UNLLNet to detect noisy labels.

EIDRM first obtains coarse difference features df_i by directly differentiating the bitemporal features of the first three feature blocks (i.e., $i = 1, 2, 3$). To fully leverage the edge prior information, parallel convolution operations with different dilation rates and padding rates are performed on df_i. This approach enables the capture of changes in different receptive fields while preserving the spatial integrity. The specific process can be formalized as follows:

$$df_i^r = f_{3\times3}\left(df_i; r\right), \quad r \in \{1, 3, 5, 7\} \quad (5)$$

where r denotes the dilation and padding rate.

Subsequently, these feature maps from different receptive fields are aggregated along the channel dimension to further enhance the spatial structure information. The edge structure is refined through the explicit integration of edge information into the difference features, thereby improving the quality of the CD results. The process can be expressed as follows:

$$df_i' = f_\sigma\left(f_{1\times1}\left(\left[df_i^1, df_i^3, df_i^5, df_i^7\right]\right)\right) \quad (6)$$

where $f_{1\times1}$ represents the 1×1 convolution block to estimate the edge information, and $f_\sigma(\cdot)$ is the sigmoid function. The edge probability map df_i' is employed to multiply and add onto the original difference-feature map df_i to realize the weight redistribution and obtain the edge-enhanced difference features EF_i. Notably, deep supervision will be applied to the intermediate edge features EDGE_i to achieve the effective edge calibration.

Nevertheless, the obtained difference features EF_i may still contain change-irrelevant edge components. To address this limitation, an ILAtt mechanism is introduced, which fosters a greater focus on the edge of change regions. Specifically, the bitemporal features F_1^{i+1}, F_2^{i+1}, and the edge-enhanced difference features EF_i are first mapped and reshaped into three feature embeddings: query Q, key K, and value $V \in \mathbb{R}^{N\times C}$. ILAtt acquires query Q from EF_i, which integrates both change and edge information, facilitating a comprehensive interaction between different levels. For the estimation of attention distribution, the self-attention mechanism is followed. Taking $A_1^{1,2} \in \mathbb{R}^{N\times N}$ attained from the first and second levels as an example, ILAtt conducts an elementwise multiplication between Q and the transpose matrix of K_1 to estimate their similarity. This similarity is treated as the weight distribution

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assigned to V_1. Mathematically, the process can be formulated as follows:

$$A_j^{i,i+1} = \Phi\left(Q^i \cdot \left(K_j^{i+1}\right)^T\right) \cdot V_j^{i+1} + F_j^{i+1} \tag{7}$$

where $\Phi(\cdot)$ represents the softmax operation. $j \in \{1, 2\}$ represents the temporal phase. $A_j^{i,i+1}$ indicates the weight redistribution of F_j^{i+1}, resulted from the interaction between features at ith level and $(i+1)$th level through ILAtt. Functionally, the interaction motivates the model to capture feature representation across different levels, efficiently propagating edge and change semantics, and it promotes the global attention to focus more on the genuinely change regions. The two difference features CF_{i} and SF_{i} are derived through concatenation and subtraction operations, respectively. The validity of obtaining the two distinct difference features will be discussed in the subsequent ablation experiments.

C. Joint Detection Strategy Based on Uncertainty Analysis

Label information plays a crucial role in RS CD. However, noisy labels, particularly prevalent at edge regions, substantially compromise the CD performance. Most existing methods employ the small-loss strategy for noisy labels detection, which fails to distinguish between hard samples and noisy samples, as they are prone to occur simultaneously in the CD task and typically exhibit large loss. To address this limitation, in this article, a joint detection strategy based on uncertainty analysis is first proposed. By jointly considering uncertainty and probability, the proposed strategy can select the most reliable samples, thereby providing a robust foundation for subsequently noisy labels detection and correction.

Given that $X = \{(x_{11}, x_{21}), (x_{12}, x_{22}), \ldots, (x_{1N}, x_{2N})\}$ represents all the pixel pairs in the given bitemporal images, $N = H \times W$ represents the number of pixel pairs, the corresponding noisy labels can be described as $\widehat{Y}_{\mathrm{cd}} = \{\widehat{y}_1, \widehat{y}_2, \ldots, \widehat{y}_N\}$. The KL divergence distance between three CD predictions and the average prediction is utilized to estimate the uncertainty map for each sample

$$U = \sum_{i=1}^{3} \mathrm{KL}\left(\mathrm{P}_{\mathrm{avg}} \left\| \mathrm{P}_{\mathrm{cd}(i)}\right.\right) = \sum_{i=1}^{3} \mathrm{P}_{\mathrm{avg}} \cdot \log\left(\mathrm{P}_{\mathrm{avg}} \big/ \mathrm{P}_{\mathrm{cd}(i)}\right) \tag{8}$$

where $\mathrm{P}_{\mathrm{avg}}$ represents the average prediction, $\mathrm{KL}(\cdot)$ denotes KL divergence, and U stands for the uncertainty map.

The computed uncertainty map indicates the confidence of each sample, where lower uncertainty suggests higher confidence, and vice versa. Based on this, the proposed joint detection strategy selects a certain proportion of samples with the lowest uncertainty as reliable samples. A probability threshold is set to further select high-confidence samples that the model is most likely to predict correctly

$$\mathrm{PT}(t) = \mathrm{PT}(t_0) - \eta(t - t_0), \quad t \geq t_0 \tag{9}$$

where $\mathrm{PT}(\cdot)$ represents the probability threshold function, which linearly varies with the current epoch t. Inspired by [48], the initialized threshold is set to 0.95. η denotes the decay coefficient. t_0 is the number of warm-up epochs. Probability is preferable for analysis compared with loss due to its fixed range of [0:1], facilitating interpretation and evaluation. The process of high-confidence sample selection can be expressed as follows:

$$\begin{aligned}\mathcal{S} &= \left\{(x_{1m}, x_{2m}; \hat{y}_m) \;\middle|\; \mathrm{prob}_m > \mathrm{PT}(t)\right\} \\ &= \{s_1, s_2, \ldots, s_m\} \\ &= \{\mathcal{C} \cup \mathcal{N}\}, \quad m \in \{1, 2, \ldots, \lfloor \varepsilon N \rfloor\}\end{aligned} \tag{10}$$

where $\mathcal{S}$ denotes the subset of selected high-confidence samples, consisting of a clean set $\mathcal{C}$ and a noisy set $\mathcal{N}$. $\varepsilon \in [0:1]$, initialized to 0.5, denotes the proportion of samples with the lowest uncertainty.

Once the most reliable prediction results are obtained, they will be compared with the corresponding noisy labels. If the labels are consistent with the predicted categories, the samples will be classified as clean labels; Otherwise, they will be categorized as noisy labels, and the remaining unselected samples will retain their original noisy labels. The detection process can be represented as follows:

$$s_m \in \begin{cases}\mathcal{C}, & \text{if } \mathrm{pre}_m = \widehat{y}_m \\ \mathcal{N}, & \text{if } \mathrm{pre}_m \neq \widehat{y}_m.\end{cases} \tag{11}$$

Given that the noisy samples have been detected, they can be easily corrected by flipping their labels, as the labels are binary in this article, either 0 or 1.

D. Noise-Robust Loss Function

Loss function serves as a fundamental component in supervised learning methods, critically influencing the model optimization and performance. Notably, widely used loss functions, such as CE loss, $L1$ loss (mean absolute error), and $L2$ loss (mean squared error), are often suboptimal in the presence of noisy labels, as they ultimately overfit to the undetected noisy labels, thereby causing a degradation of CD performance [40], [41]. To further mitigate the negative impact of undetected noisy labels, in this article, a noise-robust loss function with adaptive hard sample enhancement for RS CD is designed. The total loss [formulated as (12)] can be decomposed into two components: CD loss and edge calibration loss. The former ensures the model outputs high-quality CD results, while the latter helps to calibrate the edge structure of the CD results, thereby further improving the performance of the proposed UNLLNet

$$L_{\mathrm{total}} = \sum_{i=1}^{3} \left(L_{\mathrm{cd}(i)} + L_{\mathrm{edge}(i)}\right) \tag{12}$$

where $L_{\mathrm{cd}(i)}$ and $L_{\mathrm{edge}(i)}$ represent CD loss and edge calibration loss, respectively.

1) For the CD loss, uncertainty is integrated into the CE loss, and the loss weights of each sample are adaptively adjusted according to their uncertainty. More precisely, higher uncertainty indicates a harder sample, resulting in a larger loss weight, while lower uncertainty suggests an easier sample with a smaller loss weight. In this way, hard samples are adaptively enhanced, enabling the model to learn from more complex and challenging feature representations and improving the model's

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robustness to noisy labels. Since all the noisy samples are assumed to be unknown, the model parameters are updated by minimizing the loss values between three CD predictions and the corrected noisy labels one by one

$$\mathrm{CE}_i^{h,w} = -\widehat{Y}_{\mathrm{cd}}^{h,w} \log \mathrm{P}_{\mathrm{cd}(i)}^{h,w} - \left(1 - \widehat{Y}_{\mathrm{cd}}^{h,w}\right) \log \left(1 - \mathrm{P}_{\mathrm{cd}(i)}^{h,w}\right) \tag{13}$$

$$L_{\mathrm{cd}(i)} = \frac{1}{H \times W} \sum_{h=1,w=1}^{H,W} \left(\left(\mathrm{CE}_i^{h,w} + U^{h,w}\right) \cdot e^{U^{h,w}}\right) \tag{14}$$

where $\mathrm{CE}_i^{h,w}$ is the CE loss for the pixel at location (h, w).

2) For the edge calibration loss, $L1$ loss is utilized to capture the edge structure of change regions according to the edge information extracted by EIDRM. Specifically, the three intermediate edge features $\mathrm{EDGE_i}$ from EIDRM are fed into the edge prediction heads to obtain corresponding edge detection maps $\mathrm{P}_{\mathrm{edge}(i)}$

$$Y_{\mathrm{edge}} = \sqrt{\left(\widehat{Y}_{\mathrm{cd}} * G_x\right)^2 + \left(\widehat{Y}_{\mathrm{cd}} * G_y\right)^2} \tag{15}$$

$$L_{\mathrm{edge}(i)} = \left\| \mathrm{P}_{\mathrm{edge}(i)} - Y_{\mathrm{edge}} \right\|_1 \tag{16}$$

where Y_{edge} represents the edge labels, generated by a sobel operator [56] from the noisy labels. G_x and G_y represent the horizontal and vertical convolution kernels of the Sobel operator, respectively.

III. Experiments

A. Datasets

RS images have become an important data source for CD tasks due to their unique advantages in large-scale environmental monitoring, which can be attributed to their synoptic coverage and multitemporal acquisition capabilities. To verify the effectiveness of the proposed UNLLNet, extensive experiments are conducted on three widely used RS CD datasets.

1) LEVIR-Cd [57]: LEVIR-CD is a very-HR (0.5 m/pixel) dataset released by Beihang University. It consists of 637 image pairs with a spatial size of 1024 × 1024. Considering the limited GPU memory capacity, each image is cropped into 16 nonoverlapping patches of 256 × 256 pixels, and they were randomly split into 7120/1024/2048 for training/validation/testing, respectively.

2) CDD [58]: CDD is a season-varying dataset acquired from Google Earth, with spatial resolutions ranging from 3 to 100 cm/pixel. It consists of 16 000 image pairs with a spatial size of 256 × 256. In this work, the image dataset was randomly divided into 10 000/3000/3000 for training/validation/test sets.

3) CLCD [59]: CLCD focuses on the farmland changes collected by Gaofen-2 in Guangdong, China. It contains 600 pairs of 512 × 512 farmland RS images, with a range of spatial resolutions from 0.5 to 2 m/pixel. Similarly, the images are cropped into nonoverlapping patches of size 256 × 256 and split them into training/validation/test sets with a ratio of 6:2:2.

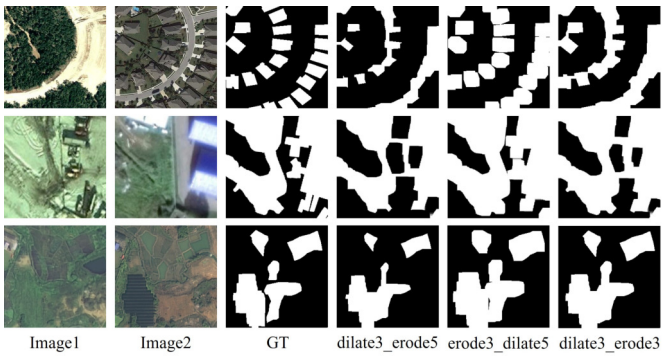


Fig. 4. Sample images from three datasets. The pre- and post-RS images are shown in the first two columns. The third column illustrates the GT. The fourth, fifth, and sixth columns present the noisy labels under dilate3_erode5, erode3_dilate5, and dilate3_erode3 settings, respectively.

All three RS CD datasets mentioned above are characterized by noise-free label annotations. To better simulate the real-world scenarios, where noisy labels are more likely to be introduced at edge regions, three types of noisy labels are implemented through dilation and erosion operations (e.g., dilate3_erode5, erode3_dilate5, and dilate3_erode3). Taking dilate3_erode5 as an example, it denotes a sequence of three dilation operations followed by five erosion operations. To investigate the robustness of the proposed UNLLNet under different noise ratios, experiments are conducted with three noise ratios: $\rho \in \{0.1, 0.3, 0.5\}$ for each noise type, where $\rho = 0.1$ indicates that 10% of the label images are subjected to noise injection. Fig. 4 illustrates several sample images from the aforementioned three RS CD datasets along with their corresponding labels. As shown in the figure, under the configured noise settings, initially discrete change targets merge into interconnected regions. The overall change area exhibits expansion or contraction, depending on the number of dilation and erosion operations performed, while edge regions demonstrate significant blurring effects.

B. Experimental Setup

1) Comparison Methods: To authenticate the superiority of UNLLNet, eight state-of-the-art methods are selected for comparison, including FC-Siam-Di [14], FC-Siam-Conc [14], BIT [17], MSCANet [19], DMINet [60], AMTNet [20], DA-IDANet [21], STFF-GA [61], and PUF [62]. These approaches have achieved remarkable performance on the RS CD task in different ways. To ensure the fairness of the experiments, their hyperparameters are configured according to the settings specified in their original literature.

2) Implementation Details: In this article, all the experiments are implemented using the PyTorch framework with a GeForce RTX 3090Ti GPU. The optimizer of the proposed network is AdamW. The initial learning rate is set to 0.0001, which decays linearly with the epochs, the weight decay is set to 0.01, the batch size is set to 8, the decay coefficient η of the probability threshold function is set to 0.002, and the number of warm-up epochs t_0 and training epochs is set to 15 and 80, respectively. Notably, the same data loading approach is employed for each method to facilitate the introduction of noisy labels. For the correction of noisy labels, they are

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TABLE II
QUANTITATIVE RESULTS ON LEVIR-CD DATASET. THE BOLD AND UNDERLINE INDICATE THE BEST AND SECOND-BEST VALUE (%)

Noise Ratio	Methods	Year	dilate3_erode5	erode3_dilate5	dilate3_erode3
			Precision / Recall / F1 / IoU / OA	Precision / Recall / F1 / IoU / OA	Precision / Recall / F1 / IoU / OA
0.1	FC-Siam-Di	2018	86.24 / 79.58 / 82.78 / 70.61 / 98.31	82.89 / 83.45 / 83.17 / 71.19 / 98.28	84.09 / 81.06 / 82.55 / 70.28 / 98.25
	FC-Siam-Conc	2018	87.14 / 78.11 / 82.38 / 70.04 / 98.30	84.72 / 80.87 / 82.75 / 70.58 / 98.28	86.79 / 78.41 / 82.39 / 70.05 / 98.29
	BIT	2022	92.53 / 82.86 / 87.43 / 77.66 / 98.79	90.50 / 84.75 / 87.53 / 77.83 / 98.77	90.87 / 85.45 / 88.08 / 78.69 / 98.82
	MSCANet	2022	92.09 / 83.07 / 87.35 / 77.54 / 98.77	89.73 / 86.12 / 87.88 / 78.39 / 98.79	88.96 / 87.02 / 87.98 / 78.53 / 98.79
	DMINet	2023	**92.59** / 85.42 / 88.86 / 79.95 / 98.91	90.27 / 87.90 / 89.07 / 80.29 / 98.90	91.27 / 87.29 / 89.23 / 80.56 / 98.93
	AMTNet	2023	90.98 / 86.98 / 88.93 / 80.07 / 98.90	90.43 / 87.38 / 88.88 / 79.98 / 98.87	**92.01** / 85.98 / 88.89 / 80.00 / 98.91
	DA-IDANet	2024	84.93 / 93.11 / 88.83 / 79.91 / 98.81	84.64 / **93.46** / 88.83 / 79.91 / 98.80	86.71 / 92.32 / 89.43 / 80.87 / 98.89
	STFF-GA	2024	89.83 / 87.20 / 88.50 / 79.37 / 98.85	88.26 / 89.26 / 88.76 / 79.79 / 98.85	88.97 / 88.84 / 88.91 / 80.03 / 98.87
	PUF	2024	85.07 / **93.91** / 89.27 / 80.62 / 98.96	88.72 / 91.20 / 89.94 / 81.72 / 98.99	86.47 / **93.44** / 89.82 / 81.52 / 99.00
	Ours	-	91.52 / 89.24 / **90.36** / **82.42** / **99.03**	**92.63** / 88.37 / **90.45** / **82.57** / **99.05**	91.38 / 89.00 / **90.18** / **82.11** / **99.01**
0.3	FC-Siam-Di	2018	85.38 / 78.18 / 81.62 / 68.95 / 98.21	79.20 / 82.13 / 80.64 / 67.56 / 97.99	83.48 / 80.32 / 81.87 / 69.30 / 98.19
	FC-Siam-Conc	2018	85.03 / 77.67 / 81.18 / 68.32 / 98.17	77.74 / 84.32 / 80.90 / 67.92 / 97.97	80.69 / 83.83 / 82.23 / 69.82 / 98.15
	BIT	2022	91.37 / 81.70 / 86.27 / 75.85 / 98.68	85.55 / 88.14 / 86.83 / 76.72 / 98.64	87.99 / 87.00 / 87.49 / 77.76 / 98.73
	MSCANet	2022	90.93 / 81.32 / 85.85 / 75.22 / 98.64	87.54 / 86.81 / 87.17 / 77.26 / 98.70	87.74 / 87.23 / 87.48 / 77.75 / 98.73
	DMINet	2023	**92.88** / 81.99 / 87.09 / 77.14 / 98.76	87.11 / 89.07 / 88.08 / 78.69 / 98.77	89.17 / 88.17 / 88.67 / 79.64 / 98.85
	AMTNet	2023	92.56 / 83.14 / 87.59 / 77.93 / 98.80	83.29 / 91.40 / 87.16 / 77.24 / 98.63	87.25 / 89.73 / 88.47 / 79.33 / 98.81
	DA-IDANet	2024	86.45 / 89.28 / 87.85 / 78.33 / 98.74	78.44 / **94.40** / 85.68 / 74.95 / 98.39	85.07 / **92.70** / 88.72 / 79.73 / 98.80
	STFF-GA	2024	88.71 / 86.45 / 87.56 / 77.88 / 98.75	83.89 / 91.23 / 87.40 / 77.63 / 98.66	86.86 / 89.73 / 88.27 / 79.01 / 98.79
	PUF	2024	83.55 / **92.72** / 87.90 / 78.41 / 98.83	90.35 / 86.20 / 88.23 / 78.93 / 98.77	88.46 / 89.84 / 89.14 / 80.41 / 98.90
	Ours	-	91.18 / 87.80 / **89.46** / **80.92** / **98.95**	**91.13** / 88.05 / **89.57** / **81.10** / **98.96**	**92.38** / 86.87 / **89.54** / **81.07** / **98.97**
0.5	FC-Siam-Di	2018	85.99 / 69.17 / 76.67 / 62.16 / 97.86	71.94 / 86.73 / 78.64 / 64.80 / 97.60	80.16 / 81.83 / 80.99 / 68.05 / 98.04
	FC-Siam-Conc	2018	86.51 / 70.31 / 77.57 / 63.36 / 97.93	74.55 / 83.36 / 78.71 / 64.89 / 97.70	79.30 / 83.43 / 81.31 / 68.51 / 98.05
	BIT	2022	90.38 / 72.26 / 80.31 / 67.10 / 98.20	79.17 / 89.73 / 84.12 / 72.59 / 98.27	84.12 / 87.87 / 85.96 / 75.37 / 98.54
	MSCANet	2022	90.43 / 73.48 / 81.08 / 68.18 / 98.25	79.19 / 90.64 / 84.53 / 73.21 / 98.31	86.80 / 88.02 / 87.41 / 77.63 / 98.71
	DMINet	2023	91.19 / 73.63 / 81.47 / 68.74 / 98.29	80.41 / 91.09 / 85.41 / 74.54 / 98.42	87.07 / 88.70 / 87.87 / 78.37 / 98.75
	AMTNet	2023	88.91 / 82.93 / 85.82 / 75.16 / 98.60	78.53 / 91.45 / 84.50 / 73.16 / 98.29	86.67 / 88.77 / 87.71 / 78.11 / 98.73
	DA-IDANet	2024	87.73 / 82.87 / 85.23 / 74.26 / 98.54	75.27 / **94.43** / 83.77 / 72.07 / 98.14	79.31 / **94.62** / 86.29 / 75.89 / 98.47
	STFF-GA	2024	87.24 / 78.56 / 82.68 / 70.47 / 98.32	76.82 / 92.60 / 83.98 / 72.38 / 98.20	82.85 / 90.78 / 86.63 / 76.42 / 98.57
	PUF	2024	76.62 / **91.40** / 83.36 / 71.46 / 98.44	**90.77** / 78.87 / 84.40 / 73.02 / 98.29	89.36 / 87.46 / 88.40 / 79.21 / 98.81
	Ours	-	**92.22** / 83.57 / **87.68** / **78.07** / **98.80**	87.70 / 89.18 / **88.43** / **79.27** / **98.81**	**90.05** / 88.93 / **89.48** / **80.97** / **98.94**

updated every 5 epochs after warm-up and stop updating at the last 15 epochs.

3) Evaluation Metrics: For the evaluation, five widely recognized metrics are utilized, including Precision, Recall, $F1$, IoU, and OA [63]. The corresponding definitions are as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{17}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{18}$$

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{19}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}} \tag{20}$$

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{21}$$

where TP, TN, FP, and FN represent the number of true positive, true negative, false positive, and false negative pixels, respectively.

C. Performance Comparison

1) Quantitative Evaluation: Tables II–IV elaborate the results of all the competing methods on three datasets, respectively. Values highlighted in bold and underlined indicate the best and second-best results.

For the LEVIR-CD dataset, taking dilate3_erode5_0.5 as an example, UNLLNet achieves the best performance across all five metrics, outperforming the AMTNet by 3.31%, 0.64%, 1.86%, 2.91%, and 0.2% in terms of Precision, Recall, $F1$, IoU, and OA, respectively. Furthermore, as observed in Table II, it is denoted that the performance of all the methods deteriorates with the increase of noise ratio. PUF achieves suboptimal performance at noise ratios of 0.1 and 0.3, demonstrating strong robustness. However, our model exhibits the smallest decline, indicating its superior robustness to noisy labels. Specifically, from noise ratio 0.1 to 0.5, the IoU generated by UNLLNet decreases 4.35%, 3.3%, and 1.14% for the three noise types: dilate3_erode5, erode3_dilate5, and dilate3_erode3, respectively, while DMINet for 11.21%, 5.75%, and 2.19%, and STFF-GA for 8.9%, 7.41%, and 3.61%, respectively.

For the CDD dataset, the UNLLNet demonstrates a relative improvement across different evaluation metrics. Taking erode3_dilate5_0.5 as an example, the DMINet achieves satisfactory performance. This is because it effectively highlights the multilevel difference features through intertemporal interaction. As indicated in Table III, the UNLLNet outperforms DMINet across all metrics. Although the Precision of UNLLNet is lower than BIT, indicating a higher number of false positives, it has the highest Recall, $F1$, IoU, and OA, which respectively surpass those of BIT by 7.82%, 4.27%, 6.93%, and 0.88%.

As for the CLCD dataset, it can be observed that the quantitative results of UNLLNet with dilate3_erode5_0.1 setting are slightly inferior to AMTNet, only achieving the highest Recall. However, with the increase in noise ratio, AMTNet exhibits a more pronounced decline. Consequently, UNLL-

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TABLE III

QUANTITATIVE RESULTS ON CDD DATASET. THE BOLD AND UNDERLINE INDICATE THE BEST AND SECOND-BEST VALUE (%)

Noise Ratio	Methods	Year	dilate3_erode5	erode3_dilate5	dilate3_erode3
			Precision / Recall / F1 / IoU / OA	Precision / Recall / F1 / IoU / OA	Precision / Recall / F1 / IoU / OA
0.1	FC-Siam-Di	2018	86.78 / 51.71 / 64.81 / 47.94 / 93.37	83.48 / 53.08 / 64.90 / 48.04 / 93.23	80.57 / 55.21 / 65.52 / 48.72 / 93.14
	FC-Siam-Conc	2018	90.24 / 52.88 / 66.68 / 50.02 / 93.77	88.34 / 54.81 / 67.65 / 51.12 / 93.82	84.73 / 57.14 / 68.25 / 51.80 / 93.73
	BIT	2022	**95.36** / 83.12 / 88.82 / 79.89 / 97.53	94.55 / 83.88 / 88.90 / 80.01 / 97.53	**94.42** / 84.95 / 89.44 / 80.89 / 97.63
	MSCANet	2022	92.12 / 78.40 / 84.71 / 73.47 / 96.66	91.02 / 82.60 / 86.60 / 76.38 / 96.99	90.59 / 84.09 / 87.22 / 77.34 / 97.09
	DMINet	2023	94.09 / 88.73 / 91.33 / 84.05 / 98.01	94.28 / 88.89 / 91.50 / 84.34 / 98.05	94.06 / 90.15 / 92.07 / 85.30 / 98.17
	AMTNet	2023	94.79 / 90.25 / 92.47 / 85.99 / 98.27	**95.36** / 87.90 / 91.48 / 84.30 / 98.07	92.72 / 93.08 / 92.90 / 86.74 / 98.32
	DA-IDANet	2024	87.80 / 93.06 / 90.35 / 82.40 / 97.66	94.09 / 88.29 / 91.10 / 83.65 / 97.96	88.35 / 92.71 / 90.47 / 82.61 / 97.70
	STFF-GA	2024	88.22 / 86.78 / 87.49 / 77.77 / 97.07	88.24 / 86.81 / 87.52 / 77.81 / 97.08	87.59 / 87.97 / 87.78 / 78.22 / 97.11
	PUF	2024	89.10 / **94.69** / 91.81 / 84.86 / 98.12	89.96 / **94.26** / 92.06 / 85.29 / 98.17	91.74 / **93.95** / 92.83 / 86.62 / 98.33
	Ours	-	93.93 / 92.00 / **92.95** / **86.83** / **98.35**	93.79 / 91.34 / **92.55** / **86.13** / **98.28**	94.07 / 92.04 / **93.04** / **86.99** / **98.37**
0.3	FC-Siam-Di	2018	85.78 / 50.84 / 63.84 / 46.89 / 93.21	82.04 / 52.68 / 64.16 / 47.23 / 93.06	80.57 / 55.21 / 65.52 / 48.72 / 93.14
	FC-Siam-Conc	2018	88.66 / 54.53 / 67.52 / 50.97 / 93.81	86.12 / 55.33 / 67.37 / 50.80 / 93.68	88.18 / 55.23 / 67.92 / 51.42 / 93.84
	BIT	2022	95.40 / 79.83 / 86.92 / 76.87 / 97.17	94.40 / 82.75 / 88.19 / 78.87 / 97.39	93.21 / 85.90 / 89.41 / 80.84 / 97.60
	MSCANet	2022	94.33 / 75.60 / 83.93 / 72.31 / 96.59	90.84 / 79.49 / 84.79 / 73.59 / 96.63	92.21 / 81.74 / 86.66 / 76.46 / 97.03
	DMINet	2023	95.66 / 84.72 / 89.86 / 81.59 / 97.74	**94.45** / 87.17 / 90.66 / 82.92 / 97.88	93.37 / 90.31 / 91.81 / 84.86 / 98.10
	AMTNet	2023	**95.82** / 86.76 / 91.06 / 83.59 / 97.99	93.26 / 87.78 / 90.52 / 82.50 / 97.56	93.22 / 91.52 / 92.36 / 85.81 / 98.21
	DA-IDANet	2024	94.61 / 86.81 / 90.54 / 82.72 / 97.86	92.73 / 87.50 / 90.04 / 81.88 / 97.72	**94.53** / 88.28 / 91.30 / 83.99 / 98.02
	STFF-GA	2024	89.54 / 83.92 / 86.64 / 76.42 / 96.95	85.90 / 86.15 / 86.02 / 75.48 / 96.70	87.66 / 86.64 / 87.15 / 77.22 / 96.98
	PUF	2024	85.65 / **94.49** / 89.85 / 81.58 / 97.72	87.37 / **93.56** / 90.36 / 82.41 / 97.80	90.43 / **93.40** / 91.89 / 85.00 / 98.12
	Ours	-	95.25 / 89.52 / **92.30** / **85.69** / **98.17**	93.59 / 90.30 / **91.92** / **85.04** / **98.16**	93.79 / 91.36 / **92.56** / **86.14** / **98.26**
0.5	FC-Siam-Di	2018	83.06 / 51.53 / 63.60 / 46.63 / 93.04	80.49 / 53.37 / 64.18 / 47.26 / 92.97	83.69 / 52.05 / 64.18 / 47.26 / 93.15
	FC-Siam-Conc	2018	85.34 / 57.01 / 68.36 / 51.93 / 93.77	86.94 / 56.02 / 68.13 / 51.67 / 93.82	86.77 / 57.62 / 69.25 / 52.96 / 93.96
	BIT	2022	95.84 / 78.12 / 86.08 / 75.56 / 97.02	**94.27** / 80.62 / 86.91 / 76.86 / 97.14	93.20 / 85.56 / 89.22 / 80.54 / 97.56
	MSCANet	2022	95.57 / 73.66 / 83.20 / 71.23 / 96.49	90.36 / 79.24 / 84.43 / 73.06 / 96.55	91.10 / 82.68 / 86.69 / 76.50 / 97.00
	DMINet	2023	96.10 / 81.66 / 88.29 / 79.03 / 97.44	93.41 / 85.81 / 89.45 / 80.91 / 97.61	92.48 / 90.77 / 91.62 / 84.54 / 98.04
	AMTNet	2023	**96.68** / 82.94 / 89.28 / 80.64 / 97.65	94.06 / 84.89 / 89.24 / 80.57 / 97.59	93.33 / 90.57 / 91.93 / 85.07 / 98.12
	DA-IDANet	2024	94.02 / 84.89 / 89.23 / 80.55 / 97.58	92.15 / 86.47 / 89.22 / 80.54 / 97.53	88.89 / 92.80 / 90.80 / 83.15 / 97.78
	STFF-GA	2024	90.95 / 81.69 / 86.07 / 75.55 / 96.88	84.60 / 85.27 / 84.93 / 73.81 / 96.43	85.76 / 87.41 / 86.58 / 76.33 / 96.80
	PUF	2024	81.24 / **95.65** / 87.86 / 78.35 / 97.35	84.07 / **94.42** / 88.94 / 80.09 / 97.53	89.83 / **93.43** / 91.59 / 84.49 / 98.05
	Ours	-	92.93 / 87.81 / **90.30** / **82.31** / **97.77**	94.10 / 88.44 / **91.18** / **83.79** / **98.02**	**93.65** / 90.64 / **92.12** / **85.39** / **98.16**

TABLE IV

QUANTITATIVE RESULTS ON CLCD DATASET. THE BOLD AND UNDERLINE INDICATE THE BEST AND SECOND-BEST VALUE (%)

Noise Ratio	Methods	Year	dilate3_erode5	erode3_dilate5	dilate3_erode3
			Precision / Recall / F1 / IoU / OA	Precision / Recall / F1 / IoU / OA	Precision / Recall / F1 / IoU / OA
0.1	FC-Siam-Di	2018	73.90 / 32.35 / 45.00 / 29.03 / 94.12	70.92 / 39.17 / 50.47 / 33.75 / 94.28	71.06 / 33.45 / 45.49 / 29.44 / 94.03
	FC-Siam-Conc	2018	68.71 / 24.33 / 35.93 / 21.90 / 93.55	69.23 / 30.02 / 41.88 / 26.49 / 93.80	71.98 / 23.04 / 34.91 / 21.15 / 93.61
	BIT	2022	70.23 / 52.83 / 60.30 / 43.16 / 94.82	69.04 / 53.98 / 60.59 / 43.46 / 94.77	70.29 / 50.59 / 58.84 / 41.68 / 94.73
	MSCANet	2022	73.30 / 71.99 / 72.64 / 57.04 / 95.97	73.21 / 67.44 / 70.21 / 54.09 / 95.74	77.01 / 66.26 / 71.23 / 55.32 / **96.02**
	DMINet	2023	71.23 / 53.82 / 61.31 / 44.21 / 94.95	71.25 / 56.60 / 63.09 / 46.08 / 95.07	69.00 / 54.93 / 61.17 / 44.06 / 94.81
	AMTNet	2023	**79.87** / 66.91 / **72.82** / **57.25** / **96.28**	74.39 / 68.09 / 71.10 / 55.16 / 95.88	72.97 / 68.16 / 70.48 / 54.42 / 95.75
	DA-IDANet	2024	79.74 / 57.90 / 67.09 / 50.48 / 95.77	71.48 / **70.45** / 70.96 / 54.99 / 95.71	**78.43** / 62.43 / 69.52 / 53.28 / 95.93
	STFF-GA	2024	61.16 / 70.08 / 65.32 / 48.50 / 94.46	61.42 / 68.87 / 64.93 / 48.08 / 94.47	70.34 / 61.77 / 65.77 / 49.00 / 95.22
	PUF	2024	54.29 / 70.13 / 61.20 / 44.09 / 94.88	51.87 / 70.36 / 59.72 / 42.57 / 94.79	55.35 / 70.44 / 61.99 / 44.92 / 94.95
	Ours	-	72.13 / **73.42** / 72.77 / 57.20 / 95.84	**76.24** / 67.86 / **71.80** / **56.01** / **95.97**	71.86 / **73.05** / **72.45** / **56.80** / 95.86
0.3	FC-Siam-Di	2018	74.70 / 31.79 / 44.59 / 28.70 / 94.12	71.88 / 36.55 / 48.46 / 31.98 / 94.22	75.30 / 30.02 / 42.93 / 27.33 / 94.06
	FC-Siam-Conc	2018	70.63 / 23.52 / 35.29 / 21.43 / 93.58	72.64 / 23.41 / 35.41 / 21.52 / 93.65	74.47 / 17.77 / 28.69 / 16.75 / 93.43
	BIT	2022	68.60 / 49.34 / 57.40 / 40.25 / 94.55	71.21 / 51.93 / 60.06 / 42.92 / 94.86	64.42 / 53.12 / 58.23 / 41.07 / 94.33
	MSCANet	2022	70.52 / 70.85 / 70.68 / 54.66 / 95.63	71.98 / 67.12 / 69.47 / 53.22 / **95.61**	72.39 / 69.18 / 70.75 / 54.73 / 95.74
	DMINet	2023	68.54 / 53.74 / 60.25 / 43.11 / 94.72	68.49 / 56.01 / 61.63 / 44.54 / 94.81	73.80 / 51.29 / 60.52 / 43.39 / 95.02
	AMTNet	2023	66.61 / **75.60** / 70.82 / 54.82 / 95.28	73.09 / 63.83 / 68.15 / 51.69 / 95.56	72.60 / 68.47 / 70.48 / 54.41 / 95.73
	DA-IDANet	2024	**76.38** / 54.87 / 63.87 / 46.91 / 95.38	**74.85** / 59.28 / 66.16 / 49.43 / 95.49	74.83 / 63.01 / 68.41 / 51.99 / 95.67
	STFF-GA	2024	60.61 / 69.86 / 64.91 / 48.05 / 94.38	58.75 / 70.39 / 64.04 / 47.11 / 94.12	63.42 / 68.29 / 65.76 / 48.99 / 94.71
	PUF	2024	47.39 / 65.11 / 54.85 / 37.79 / 94.20	51.52 / 68.69 / 58.88 / 41.72 / 94.65	55.02 / **69.80** / 61.54 / 44.44 / 94.88
	Ours	-	70.40 / 73.53 / **71.93** / **56.17** / **95.83**	68.47 / **73.02** / **70.67** / **54.65** / 95.41	**76.70** / 66.68 / **71.34** / **55.45** / **96.01**
0.5	FC-Siam-Di	2018	73.14 / 31.03 / 43.58 / 27.86 / 94.02	71.22 / 32.97 / 45.07 / 29.09 / 94.02	76.52 / 26.64 / 39.52 / 24.63 / 93.93
	FC-Siam-Conc	2018	69.69 / 22.70 / 34.24 / 20.66 / 93.51	70.60 / 20.35 / 31.60 / 18.76 / 93.44	76.12 / 14.76 / 24.73 / 14.11 / 93.31
	BIT	2022	68.43 / 48.24 / 56.58 / 39.45 / 94.49	74.49 / 47.28 / 57.84 / 40.69 / 94.87	73.82 / 46.22 / 56.85 / 39.71 / 94.78
	MSCANet	2022	78.65 / 64.08 / 70.62 / 54.59 / **96.03**	70.37 / **66.88** / 68.58 / 52.18 / 95.44	68.06 / **72.85** / 70.37 / 54.29 / 95.43
	DMINet	2023	71.46 / 49.00 / 58.14 / 40.98 / 94.75	73.31 / 50.18 / 59.58 / 42.43 / 94.93	71.11 / 52.55 / 60.44 / 43.31 / 94.88
	AMTNet	2023	71.78 / 65.93 / 68.73 / 52.36 / 95.54	74.08 / 59.79 / 66.17 / 49.45 / 95.45	**79.93** / 59.90 / 68.48 / 52.07 / 95.76
	DA-IDANet	2024	**79.57** / 50.00 / 61.41 / 44.31 / 95.32	74.78 / 58.84 / 65.86 / 49.10 / 95.46	77.45 / 63.07 / 69.52 / 53.28 / 95.83
	STFF-GA	2024	67.45 / 62.25 / 64.74 / 47.87 / 94.96	59.11 / 66.62 / 62.64 / 45.60 / 94.09	59.58 / 66.82 / 62.99 / 45.98 / 94.16
	PUF	2024	46.46 / 63.92 / 53.81 / 36.81 / 94.06	54.66 / 63.01 / 58.54 / 41.38 / 94.24	48.28 / 61.99 / 54.29 / 37.26 / 93.95
	Ours	-	69.23 / **73.10** / **71.11** / **55.17** / 95.68	**75.71** / 64.67 / **69.76** / **53.56** / **95.76**	74.45 / 66.84 / **70.44** / **54.36** / **95.89**

Net regains the best performance under dilate3_erode5_0.3 and dilate3_erode5_0.5. Among all the methods compared, UNLLNet generally achieves the best or second-best quantitative values across three datasets, demonstrating its superior and robust performance under different noise settings.

2) Qualitative Evaluation: The visualization results of competing methods on three datasets are shown in Figs. 5–7. To provide a more intuitive demonstration of the performance across different methods, in these figures, TPs, TNs, FNs, and FPs are marked in white, black, red, and green, respectively.

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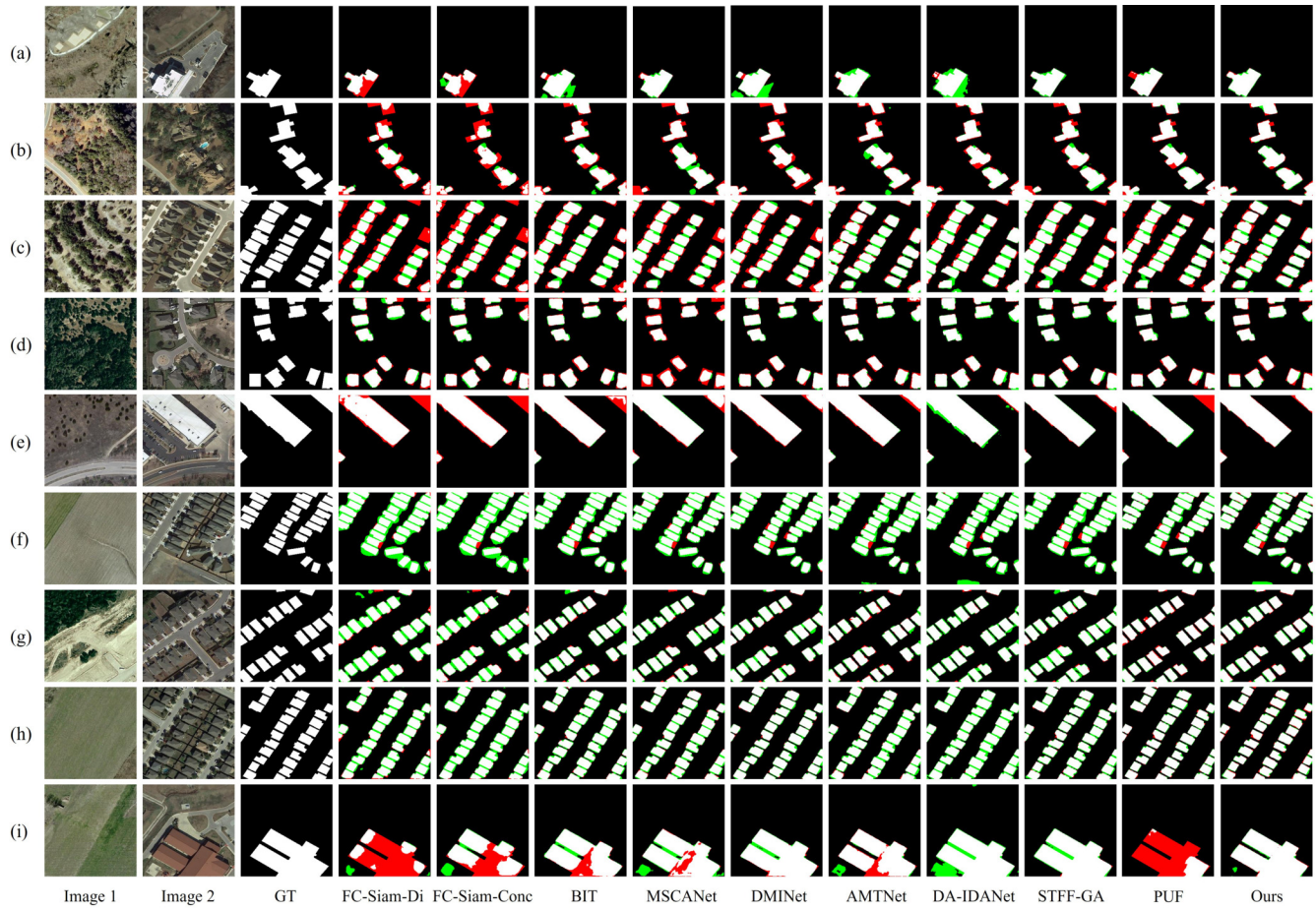


Fig. 5. Visualization results on LEVIR-CD dataset. Each row represents a noise setting with a different type or ratio. (a) dilate3_erode5_0.1. (b) dilate3_erode5_0.3. (c) dilate3_erode5_0.5. (d) erode3_dilate5_0.1. (e) erode3_dilate5_0.3. (f) erode3_dilate5_0.5. (g) dilate3_erode3_0.1. (h) dilate3_erode3_0.3. (i) dilate3_erode3_0.5. White indicates TPs. Black indicates TNs. Red indicates FNs. Green indicates FPs.

Fig. 5 presents the visualization results on the LEVIR-CD dataset. As shown in the figure, most of the competing methods are less satisfactory at detecting edges, while UNLLNet detects better edge structure of changes. When the noise ratio is set to 0.1, the impact on the comparative methods is relatively negligible, allowing them to generate satisfactory CD results. However, when the noise ratio is set to 0.5, these methods struggle to accurately locate change regions. Taking dilate3_erode5_0.5 as an example, due to the dilation operation, the competing methods erroneously classify the unchanged pixels between building objects as changed, whereas the genuinely changed pixels on either side are deemed as unchanged owing to the erosion operation. In contrast, the change map generated by UNLLNet has been improved, revealing the gaps between building objects. Such contrast is more distinct under dilate3_erode3_0.3 setting, PUF, and the proposed UNLLNet demonstrate improved CD results.

The visualization results on the CDD dataset are presented in Fig. 6. As shown in the figure, under the interference of dilate3_erode5_0.3, only our UNLLNet and PUF can still detect every single road, maintaining high identification and low false alarms. For the noise settings of dilate3_erode5_0.5, erode3_dilate5_0.3, and dilate3_erode3_0.5, similar to the observations on the LEVIR-CD dataset, other comparative methods fail to delineate the gaps between buildings. Instead, these approaches amalgamate multiple building targets into a single entity, thereby compromising the integrity of each building. However, UNLLNet successfully identifies them and significantly reduces the pseudo changes. Furthermore, UNLLNet provides a better fit to the edge structure of altered areas. These observations suggest the effectiveness of UNLLNet in highlighting relevant changes and suppressing interference from noisy labels.

Qualitative results on the CLCD dataset reveal the significant improvements of UNLLNet over other competing methods. As depicted in Fig. 7, previous methods often generate extensive error detection areas under all noise settings, which suggests their vulnerability to the noisy labels. In addition, it is noted that all the comparative methods struggle with detecting small objects effectively due to the interference of noisy labels. Taking erode3_dilate5_0.1 as an example, only AMTNet, PUF, and UNLLNet can relatively better identify the genuine changes. As for the dilate3_erode3_0.5, UNLLNet can output a more comprehensive representation of changes, while other approaches fail to capture the region on the right side. It is evident that the visualiza-

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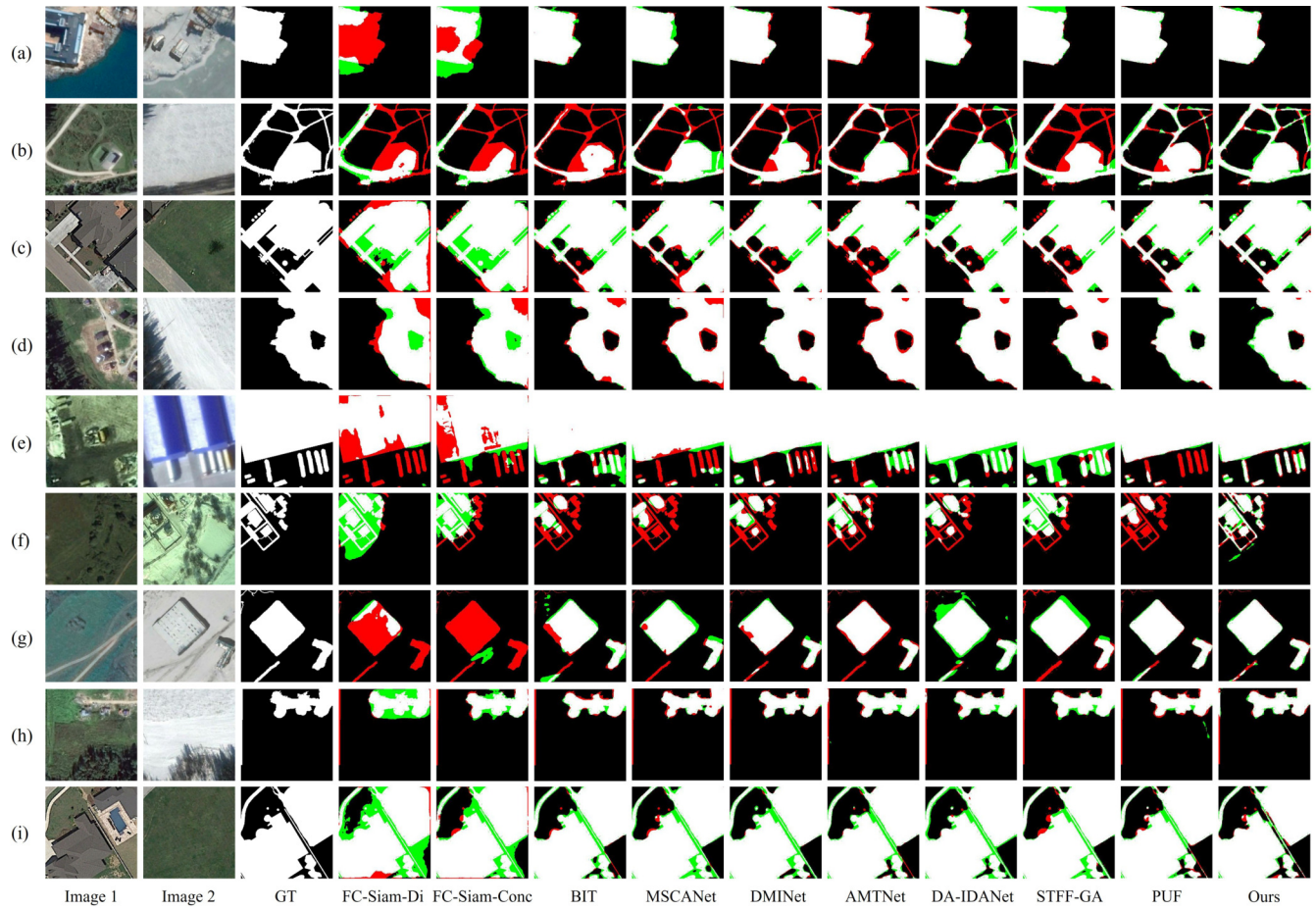


Fig. 6. Visualization results on CDD dataset. Each row represents a noise setting with a different type or ratio. (a) dilate3_erode5_0.1. (b) dilate3_erode5_0.3. (c) dilate3_erode5_0.5. (d) erode3_dilate5_0.1. (e) erode3_dilate5_0.3. (f) erode3_dilate5_0.5. (g) dilate3_erode3_0.1. (h) dilate3_erode3_0.3. (i) dilate3_erode3_0.5. White indicates TPs. Black indicates TNs. Red indicates FNs. Green indicates FPs.

tion results of UNLLNet are superior to other competing methods.

3) Efficiency Evaluation: To analyze the computational efficiency of our proposed UNLLNet, the number of parameters (Params.), FLOPs, inference time with input image sizes of 256×256, and training time on LEVIR-CD dataset for all methods are presented in Table V. It can be observed that our proposed method incurs a slight increase in computational resources compared with other methods. However, this negligible sacrifice leads to improved CD performance as well as enhanced model robustness to noisy labels, which can be accepted.

TABLE V
EFFICIENCY EVALUATION FOR ALL COMPARED METHODS ON LEVIR-CD DATASET

Methods	Params. (M)	FLOPs (G)	Inference time (s)	Training time (h)
FC-Siam-Di	1.350	4.727	0.008	1.472
FC-Siam-Conc	1.546	5.331	0.007	1.514
BIT	3.496	10.633	0.014	4.567
MSCANet	16.421	14.802	0.018	3.111
DMINet	6.242	14.551	0.012	7.167
AMTNet	24.665	21.559	0.031	3.333
DA-IDANet	16.416	49.346	0.026	6.750
STFF-GA	22.365	11.668	0.011	5.938
PUF	40.471	29.102	0.029	15.413
Ours	25.164	49.242	0.024	11.844

D. Ablation Experiments

In this section, comprehensive ablation experiments are conducted on three datasets to validate the effectiveness of our proposed UNLLNet.

1) Effectiveness of EIDRM: In UNLLNet, the EIDRM is critical to extract edge semantics and inject them into different features in an interactive manner. To confirm the effectiveness of EIDRM, relevant ablation experiments are designed, and the results under the noise setting of dilate3_erode5_0.5 are presented in Table VI (similar experimental results can be observed under other noise settings). Taking the LEVIR-CD dataset as an example, it is noted that when EIDRM is removed, the CD performance incurs a significant decline, with its Precision, Recall, $F1$, IoU, and OA decreased by 3.5%, 1.23%, 2.27%, 3.53%, and 0.09%, respectively. This indicates the EIDRM can effectively leverage the semantic information in bitemporal images, thereby improving the quality of CD results. Furthermore, the EIDRM generates two types of difference features through concatenation and subtraction operations, which are subsequently fed into the decoder and then fused together. To verify that both two different features are beneficial to our model, the impact of retaining only one type of difference feature is investigated. As shown in Table VI,

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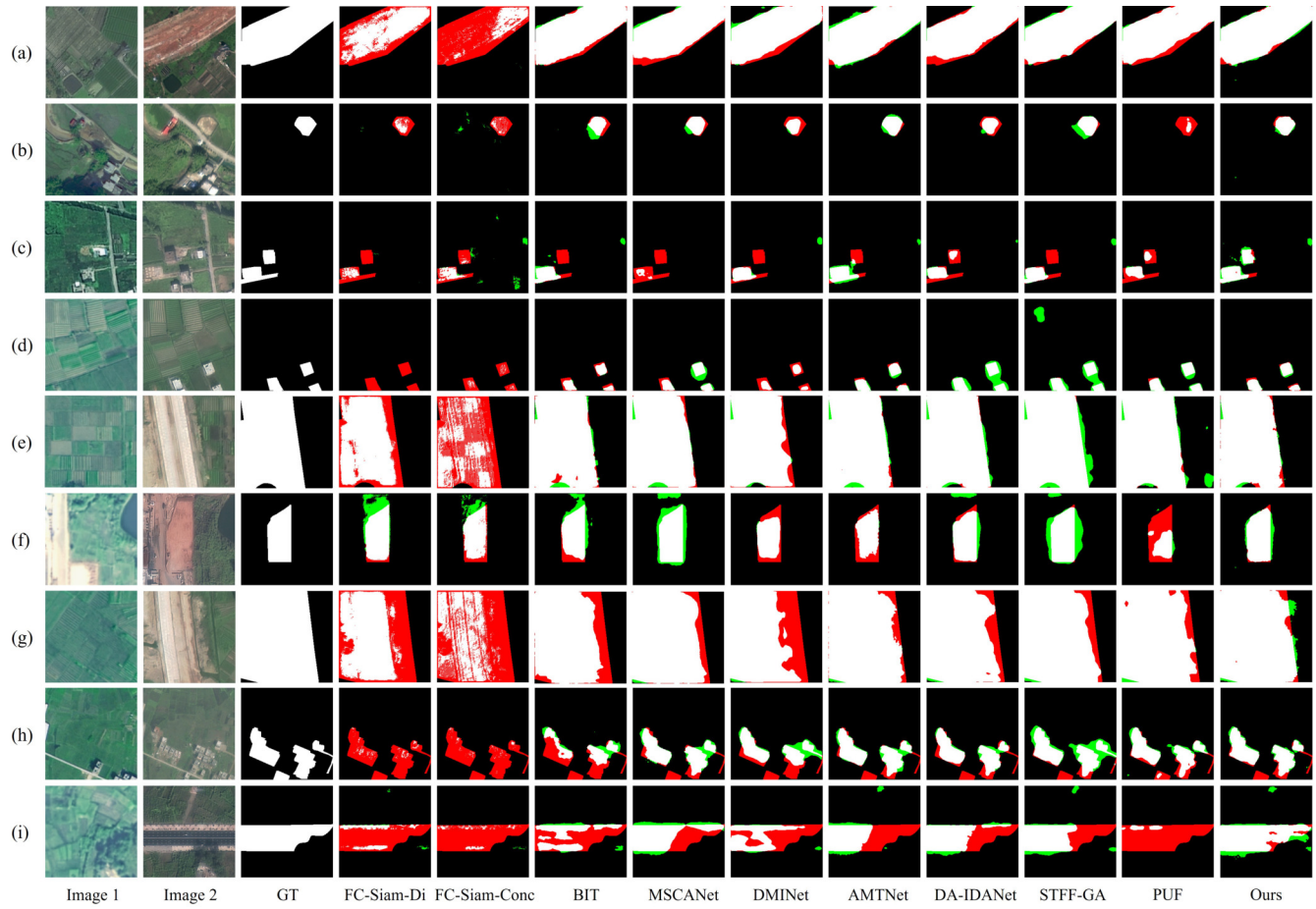


Fig. 7. Visualization results on CLCD dataset. Each row represents a noise setting with a different type or ratio. (a) dilate3_erode5_0.1. (b) dilate3_erode5_0.3. (c) dilate3_erode5_0.5. (d) erode3_dilate5_0.1. (e) erode3_dilate5_0.3. (f) erode3_dilate5_0.5. (g) dilate3_erode3_0.1. (h) dilate3_erode3_0.3. (i) dilate3_erode3_0.5. White indicates TPs. Black indicates TNs. Red indicates FNs. Green indicates FPs.

TABLE VI
ABLATION RESULTS OF EIDRM ON THREE DATASETS (%)

Dataset	EIDRM / Only Cat / Only Sub	× / × / ×	√ / √ / ×	√ / × / √	√ / √ / √
LEVIR-CD	Precision	88.72	86.51	86.26	**92.22**
	Recall	82.34	**87.81**	**87.81**	83.57
	F1	85.41	87.15	87.03	**87.68**
	IoU	74.54	77.23	77.04	**78.07**
	OA	98.71	98.68	98.67	**98.80**
CDD	Precision	93.92	93.80	**94.19**	92.93
	Recall	83.73	86.79	85.53	**87.81**
	F1	88.53	90.16	89.65	**90.30**
	IoU	79.42	82.08	81.25	**82.31**
	OA	97.44	97.76	97.67	**97.77**
CLCD	Precision	66.17	74.75	**76.45**	69.23
	Recall	71.91	66.48	64.76	**73.10**
	F1	68.92	70.37	70.12	**71.11**
	IoU	52.58	54.29	53.99	**55.17**
	OA	95.17	95.84	**95.89**	95.68

when only one difference feature is retained, whether obtained through concatenation or subtraction, the performance of the corresponding model shows a slight decline. By contrast, preserving both difference features in EIDRM leads to the best experimental results.

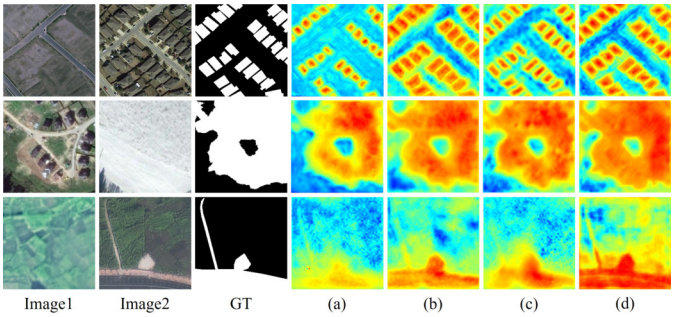


Fig. 8. Visualization of feature maps of the model with and without EIDRM. The images presented in the first, second, and third rows are derived from LEVIR-CD, CDD, and CLCD datasets, respectively. (a) Without EIDRM. (b) With EIDRM, but only cat. (c) With EIDRM, but only sub. (d) With EIDRM.

In order to more intuitively demonstrate the effectiveness of the proposed module, the feature maps of several sample images from three datasets are visualized in Fig. 8. The visualization results indicate that the model performs poorly after removing EIDRM, as it fails to extract sufficient semantic

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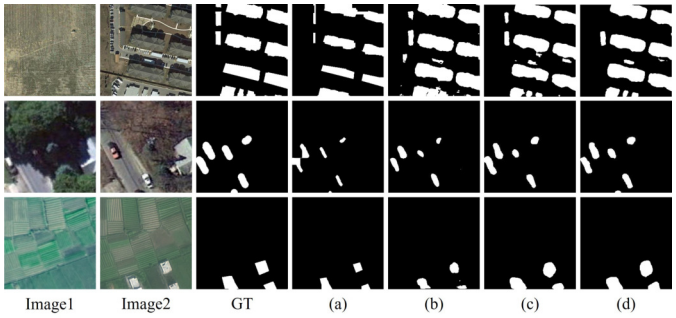


Fig. 9. Visualization of the label correction process. The images presented in the first, second, and third rows are derived from LEVIR-CD, CDD, and CLCD datasets, respectively. (a) Noisy labels. (b)–(d) Label images from different periods, where (b) epoch = 25, (c) epoch = 45, and (d) epoch = 65.

TABLE VII
ABLATION RESULTS OF JOINT DETECTION STRATEGY ON THREE DATASETS (%)

Dataset	Noise Ratio	0.1		0.3		0.5	
	Strategy	×	√	×	√	×	√
LEVIR-CD	Precision	**92.50**	91.52	86.58	**91.18**	86.23	**92.22**
	Recall	86.13	**89.24**	**88.06**	87.80	**83.79**	83.57
	F1	89.20	**90.36**	87.31	**89.46**	85.00	**87.68**
	IoU	80.51	**82.42**	77.48	**80.92**	73.91	**78.07**
	OA	98.94	**99.03**	98.70	**98.95**	98.49	**98.80**
CDD	Precision	93.25	**93.93**	90.99	**95.25**	89.04	**92.93**
	Recall	91.54	**92.00**	**90.92**	89.52	**88.93**	87.81
	F1	92.39	**92.95**	90.95	**92.30**	88.98	**90.30**
	IoU	85.86	**86.83**	83.41	**85.69**	80.16	**82.31**
	OA	98.22	**98.35**	97.87	**98.17**	97.40	**97.77**
CLCD	Precision	**72.19**	72.13	65.14	**70.40**	62.31	**69.23**
	Recall	71.80	**73.42**	**76.27**	73.53	72.35	**73.10**
	F1	71.99	**72.77**	70.27	**71.93**	66.96	**71.11**
	IoU	56.24	**57.20**	54.16	**56.17**	50.33	**55.17**
	OA	**95.84**	**95.84**	95.20	**95.83**	94.69	**95.68**

information to deal with the label noise. However, when concatenation and subtraction are embedded into EIDRM, the model can effectively calibrate the edge structure of change regions.

2) Effectiveness of Joint Detection Strategy: A joint detection strategy based on uncertainty analysis is introduced in this article to verify its capability in identifying noisy labels. Several experiments, detailed in Table VII are conducted. Specifically, the proposed strategy is intentionally omitted to systematically evaluate model performance under dilate3_erode5 for noise ratios 0.1, 0.3, and 0.5. The results reveal that our joint detection strategy can lead to improved performance under various noise settings. This phenomenon can be attributed to the fact that a model without a joint detection strategy eventually overfits to the noisy labels. Fig. 9 shows the label changes during the correction process; it is obvious that the initial noisy labels gradually converge to the true labels as the correction progresses. Although there are inevitably some erroneous corrections and undetected noisy labels, the benefits gained from accurate ones significantly outweigh this issue.

3) Effectiveness of Noise-Robust Loss Function: To assess the effectiveness and generality of introducing uncertainty into the loss function, its impact on CE loss is investigated as well as two competing loss functions (e.g., $L1$ loss, and $L2$ loss). Table VIII reports the experiment results. It can be seen that with the introduction of uncertainty, the performance of each loss function is improved, and the improvement is particularly pronounced on the $L2$ loss, which exhibits heightened sensitivity to noise due to the squaring of error. These benefits occur because the model can learn from more complex feature representations by emphasizing the analysis of uncertainty. Specifically, higher uncertainty corresponds to the more challenging samples that are harder to learn from, thereby enhancing the resistance to label noise of UNLLNet. Furthermore, CE loss achieves the best results among the three loss functions considered, which has been selected as the optimization criterion for our model. Overall, the noise-robust loss function excels by integrating the uncertainty of each sample, achieving adaptive hard sample enhancement.

4) Discussion of Initialized Threshold: Inspired by Zhu et al. [48], the initialized threshold is set to 0.95. To further investigate whether it is the best value for this article, a systematic experiment under the noise setting of

TABLE VIII
ABLATION RESULTS OF NOISE-ROBUST LOSS FUNCTION ON THREE DATASETS (%)

Dataset	Loss	CE		L1		L2	
	uncertainty	×	√	×	√	×	√
LEVIR-CD	Precision	89.76	**92.22**	**90.58**	89.97	79.79	**83.58**
	Recall	**85.43**	83.57	83.87	**85.21**	**83.49**	82.58
	F1	87.54	**87.68**	87.09	**87.53**	81.60	**83.21**
	IoU	77.84	**78.07**	77.14	**77.82**	68.92	**71.25**
	OA	98.76	**98.80**	98.73	**98.76**	98.08	**98.30**
CDD	Precision	**94.55**	92.93	**95.19**	93.03	**92.96**	91.77
	Recall	86.02	**87.81**	80.57	**85.21**	72.94	**77.99**
	F1	90.08	**90.30**	87.27	**88.95**	81.74	**84.32**
	IoU	81.95	**82.31**	77.41	**80.10**	69.12	**72.89**
	OA	**97.77**	**97.77**	97.23	**97.50**	96.16	**96.58**
CLCD	Precision	67.79	**69.23**	65.25	**72.46**	**68.18**	56.99
	Recall	72.88	**73.10**	**69.35**	65.42	56.01	**73.15**
	F1	70.24	**71.11**	67.24	**68.76**	61.50	**64.07**
	IoU	54.14	**55.17**	50.65	**52.39**	44.41	**47.13**
	OA	95.41	**95.68**	94.97	**95.58**	**94.78**	93.89

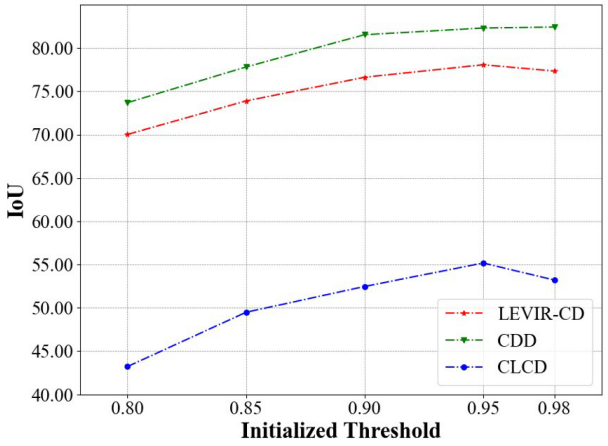


Fig. 10. Effects of varying initialized threshold from 0.8 to 0.98 on IoU.

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dilate3_erode5_0.5 is conducted by varying the initialized threshold from 0.8 to 0.98, evaluated via IoU. As indicated in Fig. 10, higher initialized thresholds lead to progressively improved IoU scores, peaking at 0.95. This phenomenon can be attributed to the fact that during the early stage of training, the model is not yet fully trained and is prone to misidentification. A lower threshold risks cumulative errors; thus, the initialized threshold is set relatively high, excluding the majority of unreliable samples. As training advances and the model's performance improves, the threshold gradually decreases, allowing for the acquisition of more high-confidence samples.

IV. Conclusion

In this article, a novel UNLLNet is proposed for RS CD, termed UNLLNet. It successfully alleviates the challenges posed by noisy labels in RS CD methods, especially at edge regions where noisy labels are more likely to be introduced. Specifically, an EIDRM is introduced to fully exploit latent edges and change semantics in bitemporal RS images. This is achieved by explicitly injecting edge information into the change features through interactions across different levels, thereby facilitating the calibration of edge structures in change regions. Then, a joint detection strategy based on uncertainty analysis is proposed, which integrate uncertainty and probability of each sample to identify noisy labels. The joint detection strategy mitigates the risk of erroneous corrections, commonly prevalent in the small-loss approach. In addition, a noise-robust loss function with adaptive hard sample enhancement is designed to further enhance the model's robustness over noisy labels. Extensive experiments on three RS CD datasets demonstrate that our proposed model achieves superior performance compared with other SOTA methods. For future work, our primary focus will be optimizing the computational efficiency of the proposed network to further reduce training time costs and enhance its practical feasibility. To this end, we will conduct rigorous validation on real-world noisy datasets to ensure robust performance across diverse scenarios. Furthermore, we aim to develop a framework with strong robustness in handling noisy labels, fully exploring intrinsic relationships among samples to achieve more accurate CD and laying a solid foundation for broader applications.

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